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AVAREBIOTECH

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Field Case Study Report on "Avare Biotech" App Validation powered by MAKSA technology on Fresh Ovine/Caprine Semen (Cordoba, Spain, 2026)

Date: January 2026
Location: Cordoba, Spain

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Executive Summary / Conclusions

- The "Avare Biotech" app is intuitive.
- Total motility measurements obtained through MAKSA were quite robust.
- Progressive motility values recorded by MAKSA were generally higher, most likely due to the system's extra detectability of progressive motion of sperms in thawed samples.
- Accurate concentration analysis requires semen videos to be recorded exactly according to Avare's guidelines. The results are highly sensitive to recording quality of input data, particularly magnification. Insufficient zoom and differences in cameras or microscopes are the primary reasons for variation in concentration values.
- The graphical demonstration of the results in a Pie chart in MAKSA report are a plus point.
- Video capture procedures followed AVARE guidelines, lead to most consistent and effective results

Profit

Using MAKSA technology, reproduction centers, insemination labs, breeding centers, and farmers can:

- **save around $26,000 per year** on CASA consumables (per 10,000 samples tested)
- **increase conception rates by ~30%** for their end clients, when checking every semen sample before AI
- **strengthen their reputation** and **attract more clients** through more consistent results.

Objective

To validate the smartphone-based MAKSA (Mobile-Assisted Key Semen Analysis) technology powering Avare Biotech App for fresh **Ovine/Caprine semen** evaluation against conventional expensive CASA systems

Materials and Methods

- Fresh semen samples from two Ovine/Caprine species were used in this study.
- The semen straws were diluted according to standard laboratory procedures

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- Following dilution, video recordings of each semen sample were captured following standardized recording conditions to minimize variability.

Table 1MAKSA results comparison

Tests done	Key Semen Parameters	Avare App	CASA	iSperm	Manual count	Difference, % (comparison with manual count)
38127 (Sheep)	Concentration M/ml	14.4	57.1	59.71	50	**247**
	Total Motility %	71	30	39	45	**57**
	Progressive Motility %	39	8	16	-	
90640 (Sheep)	Concentration M/ml	67.2	29.7	32.2	50	**34**
	Total Motility %	46	18	23	20	**130**
	Progressive Motility %	37	4	10	-	

Discussion / Expert Insights

- **MAKSA results correlate with CASA readings considering all the technical limitations and challenges Avare Biotech is dealing with.** Progressive motion sensitivity by the AI based algorithm works robustly which needs further calibrations.
- Concentration values vary and are dependent on the video magnification.
- The system sometimes showed incompatibility with **phase contrast** microscopic recordings that is likely due to contrast differentiation between sperm and the background.
- Overall, MAKSA proved intuitive, reliable, and user-friendly experience.

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Recommendations

- Standardize dilution, slide preparation, volume of drop, and temperature to minimize concentration variables and background fluid movement for sheep and goat semen samples.
- To improve users' journey it's recommended to make videos inside the app under the supervision of the app and provide guided error prompts for focus, dilution, lighting, and recording mode.

Appendix

• Screenshot of the semen videos made
• MAKSA PDF report samples

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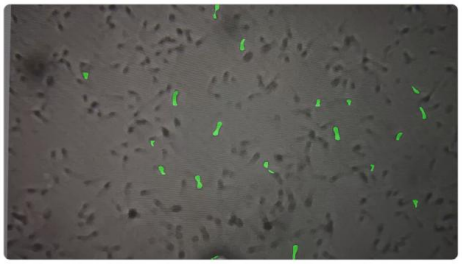
Sheep 1: Semen video

Sheep 1: MAKSA values

Analysis 17.12.2025

Animal: — **Name:** 38127 final **Age:** 5 **Breed:** Montesina **Status:** Ready **Created:** 2025-12-17 10:34



Total objects:	9
Concentration (millions/ml^3):	14.40
Dilution rate:	[1:8]
Volume of field of view (mm^3):	0.001204
Progressive Motility:	39%
Non-Progressive Motility:	32%
Immotility:	29%

Agglutination

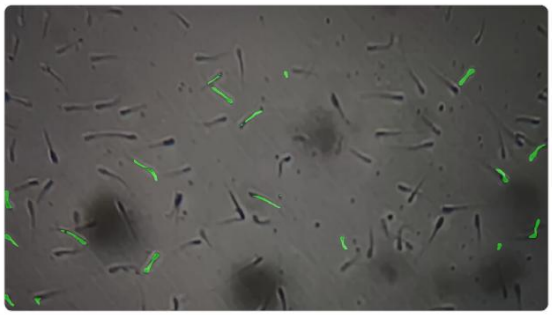
Found:	No

Sheep 2: Semen Video

Sheep 2: MAKSA values

Analysis 17.12.2025

Animal: — **Name:** 90640 **Age:** 6 **Breed:** Montesina **Status:** Ready **Created:** 2025-12-17 10:54



Total objects:	42
Concentration (millions/ml^3):	67.20
Dilution rate:	[1:24]
Volume of field of view (mm^3):	0.001863
Progressive Motility:	37%
Non-Progressive Motility:	9%
Immotility:	54%

Agglutination

Found:	No